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Dynamics and rheology of vesicles under confined Poiseuille flow

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The rheological behavior and dynamics of a vesicle suspension, serving as a simplified model for red blood cells, are explored within a Poiseuille flow under the Stokes limit. Investigating vesicle response has led to the identification of novel solutions that complement previously documented forms like the parachute and slipper shapes. This study has brought to light the existence of alternative configurations, including a fully off-centered form and a multilobe structure. The study unveils the presence of two distinct branches associated with the slipper shape. One branch arises as a consequence of a supercritical bifurcation from the symmetric parachute shape, while the other emerges from a saddle-node bifurcation. Notably, the findings are represented through diagrams that display data collapsing harmoniously based on a combination of independent dimensionless parameters. Delving into the rheological implications, a remarkable observation emerges: the normalized viscosity (*i.e.* similar to intrinsic viscosity) exhibits a non-monotonic trend as a function of vesicle concentration. Initially, the normalized viscosity diminishes as the concentration increases, followed by a subsequent rise at higher concentrations. Noteworthy is the presence of a minimum value in the normalized viscosity at lower concentrations, aligning well with the concentrations observed in microcirculation scenarios. The intricate behavior of the normalized viscosity can be attributed to a delicate spatial arrangement within the suspension. Importantly, this trend echoes the observations made in a linear shear flow scenario, thereby underscoring the universality of the rheological behavior for confined suspensions.

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1 Introduction

Approximately half the volume of blood is occupied by red blood cells (RBCs), with other constituents like white cells and platelets collectively constituting less than 1%. As a result, the flow properties of blood are predominantly governed by the dynamics of RBCs and their interactions. These interactions have diverse origins, including hydrodynamics, direct interactions (like RBC–RBC bridging *via* macromolecules such as fibrinogen), and purely entropic forces.[1,2]

RBCs are often modeled as capsules, membranes that possess in-plane shear elasticity (due to the cytoskeleton) and bending, with inextensible membranes.[3–7] An alternative model, the vesicle, represents a closed bilayer phospholipid membrane devoid of a cytoskeleton.[8–12] This vesicle model serves as a simplified representation of RBCs, enabling the discernment of whether the phenomena observed are robust or dependent on specific factors. For instance, vesicles and RBCs (modeled as capsules) share similar shapes under flow, such as the parachute and slipper forms, indicating a secondary role for the cytoskeleton.[13,14]

In some cases, dimension reduction is employed for computational efficiency, considering the membrane as a contour rather than a surface. Indeed, consider a 2D vesicle discretized into N points with mesh size Δx. To maintain the same mesh size, a 3D vesicle with triangulated mesh contains $\sim N^2$ points. When using a boundary integral method, the complexity of the simulation is $O(N^2)$ for 2D and $O(N^4)$ for 3D. Fortunately, many 2D phenomena are also reported in 3D, further highlighting their robustness. Noteworthy examples include parachute, slipper, multilobe shapes, and aggregated vesicle doublets, all observed in both 2D and 3D simulations.[7,15–20] The agreement between 2D and 3D simulations extends to studies of optimal cell transport, rheology of confined suspensions, flow-induced ATP release by RBCs, and margination of white blood cells, underscoring the generality of these behaviors.[21–25] Moreover, 2D simulations can help identify whether or not a given phenomenon is a consequence of dimensionality, or is it rather

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a robust feature. For example, multilobe shapes have been thought to result from membrane shear elasticity in 3D.[7] A recent work[18] reported similar multilobe shapes using 2D vesicle models without shear elasticity.

Studies of a single vesicle in 2D[15] unbounded Poiseuille flow have shown that the symmetric parachute solution becomes unstable below a certain flow strength, leading to a slipper shape. This scenario has been verified in 3D simulations as well.[26] Introducing walls in a 2D simulation results in a complex influence of wall effects.[16] For example, under specific conditions, chaotic dynamics emerge through a series of vesicle shape oscillations.[27] Factors like flow strength, confinement, and vesicle deflation contribute to the nuanced dynamics observed.[15,26,28]

The primary objective of this study is to delve deeper into the bifurcation structure governing various shapes and dynamics by systematically exploring a range of parameters, including flow strength, confinement, and viscosity contrast (ratio between the encapsulated viscosity over that of the suspending fluid). We present compelling evidence that there exist two distinct slipper branch solutions. One emerges as a supercritical (pitchfork) bifurcation originating from the symmetric parachute configuration, while the other materializes as a saddle-node bifurcation. This second slipper solution coexists harmoniously with both the parachute and the initial slipper solutions. Moreover, the exploration reveals the emergence of additional shapes such as multilobes, which were recently reported in 3D simulations for capsules,[7,17] suggesting the necessity of cytoskeleton elasticity. However, our findings emphasize the insignificance of shear elasticity, as we observe similar solutions for vesicles in Poiseuille flow. This conclusion is consistent with recent studies on 2D vesicles under linear shear flow,[18] where comprehensive investigations of multilobe solutions were conducted.

Another pivotal objective of our work is to demonstrate that the bifurcation diagrams can be encapsulated in universal forms, amalgamating diverse dimensionless parameters into a single universal parameter. In addition, we systematically analyze the rheological properties of both dilute and more concentrated suspensions. Notably, we reveal that the normalized viscosity $[\eta] = (\eta_{\text{eff}} - \eta_{\text{out}})/(\eta_{\text{out}}\phi)$, where η_{eff} represents the effective viscosity of the suspension and η_{out} stands for the viscosity of the suspending fluid, exhibits a decreasing trend with increasing vesicle concentration ϕ within the range of approximately 0% to 15%. This concentration interval aligns with microcirculation conditions. This intriguing behavior stems from a delicate spatial organization within the suspension.

The organization of the article is as follows. We present the model and simulation methodology in Section 2. The core findings and their discussions are housed in Section 3. Within this section, we extensively explore the effect of confinement and flow strength across a wide range of viscosity contrast values (λ = 1–10) on the lateral position of individual vesicles, unraveling their implications on rheology. Additionally, we investigate the influence of vesicle concentration on suspension rheology. In the final section, Section 4, we provide a conclusive summary of our findings and offer insights into potential future research directions.

2 Models and methods

This study was carried out using a vesicle model with a 2D configuration. We will see that some shapes and dynamics that were suspected to be due to the cytoskeleton are in fact reproduced here in 2D, ruling out the hypothesis of relevance of the cytoskeleton.

2.1 Mathematical model

We consider one or more vesicles immersed in a periodic long straight channel, defined by a box $\Omega = [-L/2, L/2] \times [-W/2, W/2]$ (where W is the width and L the length). A quadratic velocity profile is imposed as

$$u_x{}^0 = u^0_{\max}\left[1 - \left(\frac{2y}{W}\right)^2\right], \quad u^0_y = 0, \tag{1}$$

where $u^0_{\max}$ is the maximum velocity at the center line $y\equiv 0$. The dynamic viscosity of interior fluid enclosed by the vesicle membrane is denoted as η_{in}, while the exterior one is denoted as η_{out}. The viscosity contrast is then defined as $\lambda = \eta_{\text{in}}/\eta_{\text{out}}$. The typical magnitude of blood plasma viscosity is $\eta_{\text{out}} \sim 10^{-3}$ Pa s, and that of cytoplasm is $\eta_{\text{in}} \sim 10^{-2}$ Pa s. The shear rate at the channel wall $\dot{\gamma}_{\text{w}} = (\partial u_x/\partial y)|_{y=-W/2} = 4u^0_{\max}/W$ has its typical physiological value in the range 100–10^4/s,[29] depending on the blood vessel size. This implies that the Reynolds number (by using RBC radius R_0 as a length scale) Re $\sim O(10^{-4})$ is small enough, so that the flow can be considered to be in the Stokes regime.

The vesicle membrane shape is represented by a closed curve $\mathbf{X}(s,t)$, where s is a curvilinear coordinate, and t represents time. The Helfrich energy[30] takes into account bending elasticity under the constraint of local membrane inextensibility:

$$H(\mathbf{X}(s)) = \frac{\kappa_{\text{b}}}{2}\oint c^2 \mathrm{d}s + \oint \zeta \mathrm{d}s \tag{2}$$

Here κ_{b} is the bending modulus of the membrane (for RBC its value is often taken as $\kappa_{\text{b}} = 3 \times 10^{-19}$ J), c is the local curvature, and ζ is a Lagrange multiplier that enforces a constant local membrane area.[31] Thanks to the functional derivative of H, the force term acting from the vesicle membrane on the fluid can be obtained as[32]

$$\mathbf{f}(s) = \kappa_{\text{b}}\left(\frac{\mathrm{d}^2 c}{\mathrm{d}s^2} + \frac{1}{2}c^3\right)\mathbf{n} - \zeta c\mathbf{n} + \frac{\mathrm{d}\zeta}{\mathrm{d}s}\mathbf{t} \tag{3}$$

Here $\mathbf{n}$ and $\mathbf{t}$ are the normal and tangential unit vector, respectively. It should be noted that membrane viscosity is not taken into account here. Indeed, the major effect of membrane viscosity is on transient dynamics.[33,34] Note also that for steady-state behaviors, membrane viscosity was reported to have similar effects as viscosity contrast λ does.[33,35,36] In this case it is possible to combine the dissipation of membrane viscosity and internal viscosity.[37] In addition, analytical studies on vesicle dynamics

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(in the small deformation theory)[38] have shown that the bulk viscosities (inside and outside viscosities) and the membrane viscosity enter the equations in an additive way, leading to a single effective viscosity. On this basis, we can say that by varying the viscosity contrast, we can mimic the effect of membrane viscosity. However, a general statement on the effect of membrane viscosity for large deformation and time-dependent shapes remains to be established.

We define the characteristic radius of a vesicle R_0 by the relation $A = \pi {R_0}^2$, which has a typical value $R_0 = 3$ μm for a RBC. The reduced area τ is defined by the ratio between vesicle area A and the area of a circle having the same perimeter (denoted as P) as the vesicle

$$\tau = \frac{A}{\pi[P/(2\pi)]^2} = \frac{4\pi A}{P^2}. \tag{4}$$

This quantity specifies the roundness of the vesicle shape, the more round the shape is, the closer its value is to 1. We fix $\tau = 0.6$ in this study, in reference to the known value of a healthy RBC (which is about 0.65).[39]

2.2 Boundary integral formulation

We use the boundary integral method (BIM)[8,40–43] to convert the Stokes equations into an integral equation for the evolving boundaries (vesicle shape dynamics). For the rheology part we will use a lattice-Boltzmann method (LBM)[44] due to its efficiency for large hematocrit. We have nevertheless run few simulations with BIM as a benchmark. Using a spectral discretization, the membrane position $\mathbf{X}$ is represented by a truncated Fourier series taking the x and y components of $\mathbf{X}$ as the real and imaginary parts respectively:

$$X_x(s) + iX_y(s) = \sum_{k=-k_{\max}}^{k_{\max}} X_k \exp\left(\frac{2\pi iks}{P}\right) \tag{5}$$

where the complex amplitudes X_k are the shape parameters of the membrane and $k_{\max}$ is the cut-off harmonic of the Fourier series. The membrane force term $\mathbf{f}(s)$ in eqn (3) is calculated from the virtual work principle by computing the variation of energy H due to a small change of X_k. The velocity at any point $\mathbf{r}$ in the simulation domain obeys an integral equation:

$$\begin{aligned} \Lambda\mathbf{u}(\mathbf{r}) &= \mathbf{u}^0(\mathbf{r}) + \frac{1}{\eta_{\text{out}}}\oint \mathbf{f}(s') \cdot \mathbf{G}(\mathbf{X}(s'), \mathbf{r}) \mathrm{d}s' \\ &+ (1-\lambda)\oint \mathbf{u}(\mathbf{X}(s')) \cdot \mathbf{T}(\mathbf{X}(s'), \mathbf{r}) \cdot \mathbf{n}(\mathbf{X}(s')) \mathrm{d}s' \\ &+ \frac{1}{\eta_{\text{out}}}\int_W \mathbf{f}_{\text{w}}(s') \cdot \mathbf{G}(s', \mathbf{r}) \mathrm{d}s' \end{aligned} \tag{6}$$

$\mathbf{G}$ is the single layer Green's function, while $\mathbf{T}$ denotes the double-layer Green's function. Here we use the Green's functions satisfying periodic boundary conditions in the x direction.[43] The contour integrals are carried out along all vesicles in the calculation domain, and the last one is along the domain boundaries, where $\mathbf{f}_{\text{w}}$ is the force due to the wall on the fluid (unknown *a priori*). The notation $\mathbf{\Lambda}$ is defined as

$$\mathbf{\Lambda}(\mathbf{r}) = \begin{cases} \lambda, & \text{if } \mathbf{r} \text{ isinside a vesicle} \\ (1+\lambda)/2, & \text{if } \mathbf{r} \text{ is on a membrane} \\ 1, & \text{if } \mathbf{r} \text{ is out side of any vesicles} \end{cases} \tag{7}$$

It should be noted that the above Green's functions do not satisfy the no-slip boundary condition at the walls, in contrast to one of our previous studies.[45] Therefore, the wall contribution to the velocity field has to be taken explicitly into account [the 4th term on the right hand side of eqn (6)]. There are several techniques to compute the wall contribution.[8,41,45–48] In the present work, the wall force $\mathbf{f}_{\text{w}}$ is calculated by solving eqn (6) for $\mathbf{r}$ located at walls whereby the left hand side of eqn (6) is set to be equal to $\mathbf{0}$. Detailed implementation can be found in Abbasi *et al.*[18]

Once having the velocity on the membrane, the evolution of vesicle shape is obtained from a simple fixed time step Euler scheme:

$$\mathbf{X}(t + \Delta t) = \mathbf{X}(t) + \mathbf{u}(t)\Delta t \tag{8}$$

When implementing eqn (8), the velocity term $\mathbf{u}(\mathbf{X}(s'))$ on the right hand side in eqn (6) takes its value at time t.

Two additional procedures were performed in order to ensure long-term stability of the simulations. First, we keep the inner volume of the vesicles fixed. Normally, fluid incompressibility and membrane impermeability should keep the inner volume of the vesicles constant. However, a small drift due to numerical error cannot be fully excluded. We compensate this drift by inflating or deflating the elastic vesicle through homogeneous normal deformation. Second, we perform a small correction of membrane positions when two vesicles approach each other too closely (considered as a collision). The correction starts to act only when the distance between the vesicles' membranes is below a certain limit δh. The correction pushes the vesicles apart, thus preventing numerical instabilities from vesicle interpenetration.

We performed several simulations with different numbers of Fourier harmonics and sampling points, as well as different time steps and critical inter-vesicle distances δh. Based on this verification and a compromise between efficiency and accuracy, each vesicle was characterized by 63 Fourier harmonics. 2048 sampling points were used to resolve the short-range hydrodynamics interactions. The velocity of the vesicle membrane was calculated at 128 sampling points. The minimal distance δh for which the vesicles were unaffected by the collision-preventing procedure was 0.005 of the vesicle radius.

The choice of the number of harmonics for walls depends on channel length L. By using harmonics from 256 to 1024, the residual wall velocity was found to be of very small fraction of velocity in the center of the channel (with relative value less than 10^{-5}, reducing to machine precision in many cases).

The problem has three dimensionless numbers: (i) the viscosity contrast

$$\lambda = \frac{\eta_{\text{in}}}{\eta_{\text{out}}}, \tag{9}$$

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(ii) the capillary number

$$Ca = \frac{\eta_{out}\dot{\gamma}_w R_0^3}{\kappa_b} = \dot{\gamma}_w \tau_c \tag{10}$$

which describes the fluid strength over the vesicle bending strength, and (iii) the degree of confinement

$$C_n = \frac{2R_0}{W}. \tag{11}$$

Here the capillary number can also be interpreted as the ratio of the characteristic time scale for vesicle deformation τ_c and the time scale of the imposed flow $1/\dot{\gamma}_w$. The ranges of these dimensionless numbers explored in this work are: $\lambda \in [1,10]$, $Ca \in [4,80]$, and $C_n \in [1/5,2/5]$. The simulation box always has its length $L = 10W$ unless otherwise specified, which is validated as being long enough to eliminate artifacts due to periodic boundary conditions.[45] We checked also that all simulation results have numerical perimeter deviations less than 0.3%, which is taken here to be as an acceptable value.

3 Results and discussion

One main novelty of this study is that we show the existence of several solutions, more or less complex, for the same set of parameters. The traditional solution in a Poiseuille flow is the parachute solution. Another solution is the slipper one[15] where the center of mass of the cell is not at the center line, and thus the overall shape is not symmetric with respect to the flow center line. In order to identify solutions which are different from the parachute one, it is natural to define the lateral position of the vesicle as a simple measure for that deviation. This deviation, denoted as $h(t)$, is defined as

$$h(t) = \frac{\oint_{mem} \mathbf{X}(t)\cdot \mathbf{y} ds}{WP} \tag{12}$$

Where $\mathbf{y}$ is the unit vector of the y (lateral) direction. h has its values ranging from -0.5 to 0.5, with 0 denoting the center line. The final lateral position can then formally be defined as

$$h_f = \lim_{T\to\infty} \frac{\int_0^T h(t)dt}{T} \tag{13}$$

In practice, the simulation time should be large enough in comparison to the intrinsic relaxation time τ_c. The initial configuration in the simulation corresponds to a well-relaxed biconcave shape aligned along the flow direction, with a zero flow field. This choice is of course arbitrary, and any other choice could be selected as well. We shall discuss later the impact of initial conditions on the results. $h_0 = h(0)$ denotes the initial lateral position. The simulation ran for long enough time until a steady, periodic, or permanently chaotic pattern is observed. Unless otherwise specified, the simulation time is $100\tau_c$. Thus, h_f can be approximated by $h_f = \int_{T_0}^{T_1} h(t)dt/(T_1 - T_0)$. Here T_0 and T_1 are chosen to guarantee that the transient effects have decayed before T_0 and that the time interval $T_1 - T_0$ is long enough to ensure the convergence of h_f.

3.1 Emergence of the off-centered final position by increasing viscosity contrast

The viscosity contrast stands out as a compelling factor exerting substantial influence over the solution behavior. To illuminate this aspect, we illustrate the primary outcome of this exploration while keeping the other two parameters $(C_n,Ca) = (2/7,80)$ constant. Through systematic adjustments of the initial position h_0 within the range of 0 to 0.4 with intervals of 0.05, we observed a dichotomy in the final positions achieved for $\lambda = 1$ and 5. As depicted in Fig. 1, the outcomes varied between a centered configuration and an off-centered one, characterized by a slipper-like shape.

The emergence of the off-centered final shape at higher values of λ is established as a robust characteristic. This behavior is explicitly portrayed in Fig. 2a, which illustrates the relationship between h_f and λ. Notably, as λ exceeds a critical value of approximately $\lambda_c^{(OC)} \approx 4$, an off-centered stable branch $(h_f^{(OC)})$ materializes. In a notable region we can aptly term the "attraction basin" (depicted in red in Fig. 2a), vesicles with initial positions and viscosity contrast pairs (h_0,λ) aligning within this area yield an off-centered final position $h_f^{(OC)}$. This specific configuration results in what is colloquially referred to as a slipper shape (as illustrated in Fig. 2b). It is worth mentioning that when $\lambda > \lambda_c^{(OC)}$, the attraction domain of $h_f^{(OC)}$ is considerably broader than that of $h_f^{(C)}$. This intriguing finding suggests that a slipper shape might be a more prevalent occurrence than the parachute shape (as shown in Fig. 2c) in physiological scenarios, given the typical range of λ around 5–10 for healthy RBCs.

This phenomenon of shape bistability, where the final shape is contingent on the initial conditions, has previously been noted in experiments and simulations using a 3D capsule model.[49,50] The current findings with the 2D vesicle model reaffirm that shape bistability is an inherent and robust trait of closed membranes. This lends further credence to the notion that the 2D vesicle model offers a dependable representation for capturing essential features.

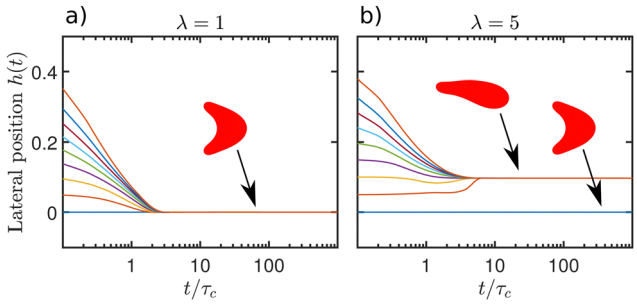


Fig. 1 (a) Normalized lateral position *versus* time. Different curves $h(t)$ correspond to different initial positions, with $\lambda = 1$. All curves converge to $h_f = 0$, which corresponds to a parachute final shape; (b) $h(t)$ curves obtained with $\lambda = 5$, where vesicles start from an off-centered initial position ($h_0 \geq 0.05$), and they reach an off-centered final position $h_f = 0.097$. If instead we start from $h_0 = 0$ we obtain $h_f = 0$ (parachute shape), which has been reported in Tahiri *et al.*[28] The stability of $h_f = 0$ is validated by imposing small perturbations about the presumed steady state solution.

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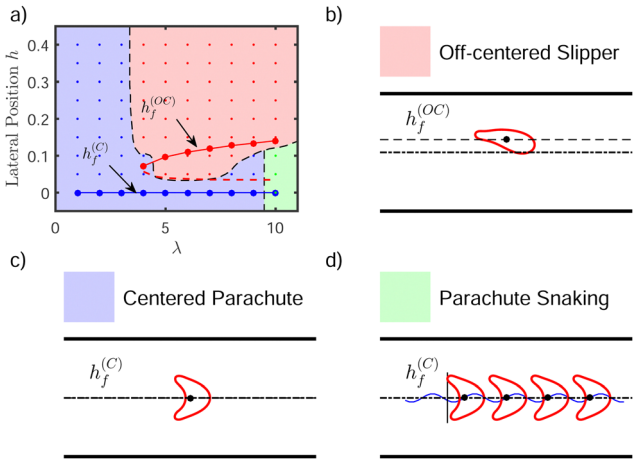


Fig. 2 The emergence of $h_f^{(OC)}$ and the corresponding final shapes, with confinement C_n = 2/7 and capillary number Ca = 80: (a) the blue and red solid lines represent the centered ($h_f^{(C)}$) and off-centered ($h_f^{(OC)}$) final position as a function of the bifurcation parameter λ. The red dashed line indicates an unstable branch, and we have added it qualitatively. Simulations are performed with different initial positions (h_0) but with a flow-aligned biconcave shape, represented by (red or blue) dots. All initial conditions within a given domain (blue or red) yield a given final position, which is either $h_f^{(C)}$ (when the initial data are in the blue region) or $h_f^{(OC)}$ (when the initial data are in the red region) depending on λ and h_0. The corresponding basins of attraction are colored in red or blue. The black dashed line represents estimated borders and are here plotted just as a guide for the eyes. Parachute snaking dynamics is observed when λ = 10 and $h_0 \leq$ 0.1. (b)–(d) Schematics for final shapes and h_f [the blue trajectory in (d)].

In Fig. 2a, the coexistence of two stable branches (indicated by the red and blue lines with filled circles) typifies the behavior of a saddle-node bifurcation. The saddle-node point is estimated from our simulations to be proximate to (h_f,λ) = (0.072,4). This estimation is corroborated by simulations with initial positions h_0 = 0.05 or 0.1, both of which lie close to h_f = 0.072 yet within the blue domain, leading to final positions centered along the midline. It is important to note that for all simulations depicted in Fig. 2a, the initial shape adopts a flow-aligned biconcave form. However, when an off-centered slipper is chosen as the initial shape (as depicted in Fig. 2b), while retaining the same initial positions (h_0 = 0.05 or 0.1), the final position unequivocally takes on an off-centered configuration $h_f^{(OC)}$. Thorough assessments have excluded potential numerical artifacts such as perimeter deviations or reduced area discrepancies. The sensitivity to the initial shape underscores the characteristics of a saddle-node bifurcation.

Beyond the off-centered slipper as a final shape, another intriguing observation materializes at λ = 10 and $h_0 \leq$ 0.1. In this scenario, a snaking-parachute shape pattern emerges (as depicted in Fig. 2d), characterized by the oscillation of the center of mass around the centerline. This snaking motion serves as an intermediate state between the parachute and another steady shape, highlighting the intricate dynamical possibilities within this system.

3.2 Effects of confinement

In addition to the influential viscosity contrast, the degree of confinement also emerges as a pivotal factor. By maintaining the capillary number Ca at 80, we present a comprehensive diagram of solutions with respect to λ for varying levels of confinement, ranging from $W = 5R_0$ to $10R_0$ (equivalent to C_n values between approximately 0.2 and 0.4), as shown in Fig. 3. Within this confinement range, Kaoui *et al.*[16] have illustrated, for λ = 1, the presence of either unconfined slipper or parachute shapes in their phase diagram. It's worth noting that for λ = 1, despite variations in vesicle shape under different confinement conditions, the final position consistently hovers close to the center.[16]

In a compelling juxtaposition with the results displayed in Fig. 2a, this new examination unveils two distinct final positions linked to two distinct final shapes (represented by the cyan and yellow regions). The cyan region denotes a domain where, upon the initial configuration falling within it, the ultimate state converges toward a near-centered final position ($h_f^{(NC)}$). The corresponding final shape closely resembles the unconfined slipper shape observed by Kaoui *et al.*[16] The transition from centered to near-centered solution corresponds to a pitchfork (continuous) bifurcation. In contrast, commencing simulations within the yellow zone leads to a complex tumbling motion characterized by apparent chaotic behavior. Extensive simulations spanning over $500\tau_c$ revealed that the final position h_f [defined in eqn (13)] oscillates in an ostensibly erratic manner between $h_f^{(NC)}$ and $h_f^{(OC)}$. Such multilobe shapes have been observed experimentally and numerically using 3D models in shear flow.[7,17] Upon systematic simulations in the vicinity of the yellow zone, we observed an abrupt transition from other stable final positions (such as slipper or parachute shapes) to chaotic behaviors. Notably, this transition to chaos is direct and does not exhibit conventional indicators of bifurcation progression towards chaos, such as subharmonic cascades or intermittency. Although uncommon, such a direct transition to chaos has been observed in simplified dynamical systems.[51]

The presence of the off-centered slipper region (highlighted in red in Fig. 3a) becomes prominent as W attains values around or beyond $5R_0$. As we systematically increase W, we observe a reduction in the critical value $\lambda_c^{(OC)}$ required for the onset of the off-centered position. This phenomenon coincides with a noticeable leftward shift of the entire red zone. This leftward migration is also observed for the cyan-colored region, which corresponds to the near-centered slipper domain and involves a reduction in the critical value $\lambda_c^{(NC)}$. The intermediate state represented by the snaking parachute (depicted in green in Fig. 3b) is aptly perceived as a transitional phase existing between the parachute and near-centered slipper configurations.

One question arises naturally whether we could distinguish between the NC and the OC solutions, as they are both off-centered and show slipper shapes. In practice, we find that it is more robust to characterize vesicle shape than lateral position, as shown in Fig. 3b. Mathematically, these two shapes can be quantified by the shape parameters X_k in eqn (5). We observed that for NC branches, $|\mathrm{Real}(X_2)| > |\mathrm{Real}(X_3)|$, whereas for OC branches, $|\mathrm{Real}(X_2)| < |\mathrm{Real}(X_3)|$. Here Real denotes the real part.

Of particular interest is the region characterized by complex tumbling-multilobe motion (designated in yellow in Fig. 3).

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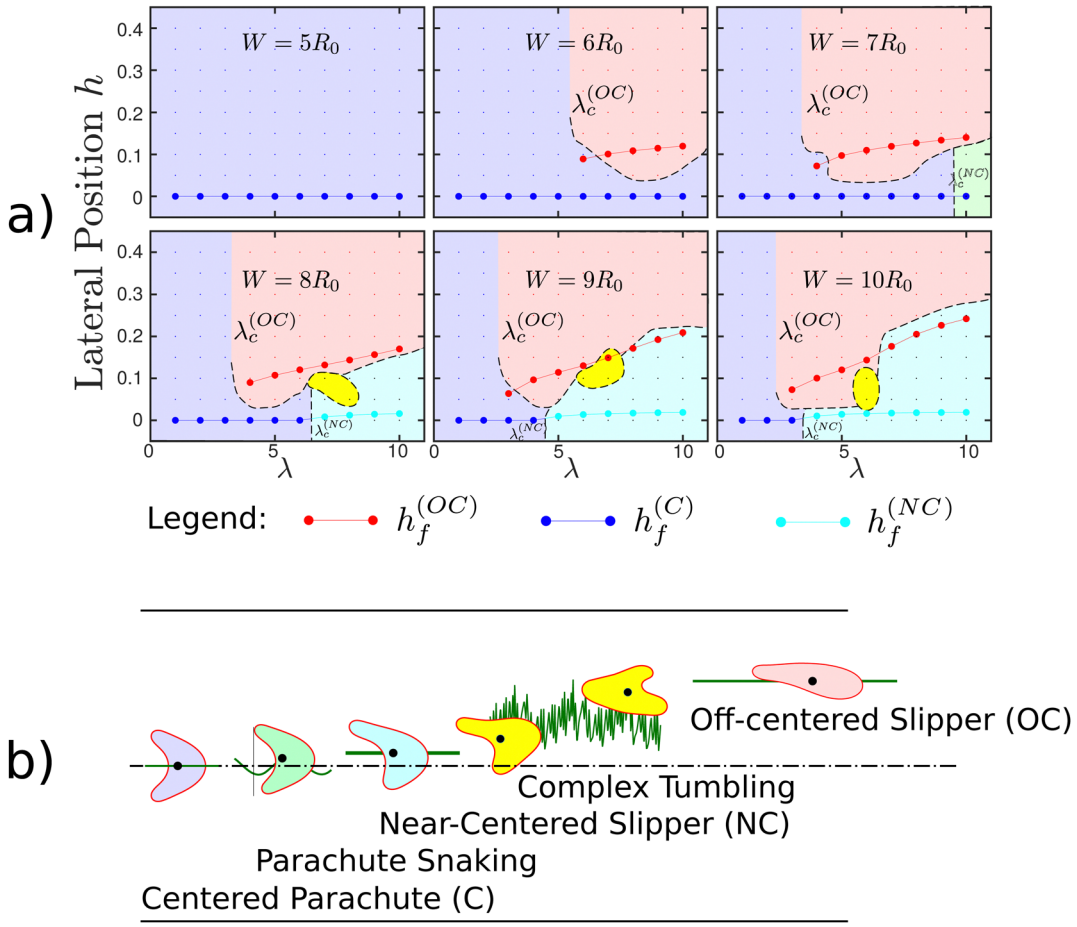


Fig. 3 (a) Evolution of final positions and final shape diagram against channel width (confinement C_n) while fixing the capillary number Ca = 80. Simulations are performed with different initial positions (h_0) but with a flow-aligned biconcave shape. The red, blue and green zones are as defined in Fig. 2a. Two distinct final patterns emerged when $W \gtrapprox 8R_0$. The yellow region represents long term tumbling (longer than $500\tau_c$ in simulation time) with apparently chaotic dynamics. The cyan region represents a final slipper shape that is slightly different from a parachute (its final position is slightly away from the center line). The dashed line represents estimated borders and are here plotted just as a guide for the eyes. The notation $\lambda_c^{(OC)}$ indicates the position of the border between red and blue regions, and $\lambda_c^{(NC)}$ is the critical value where transition from parachute to near-centered slipper happens. (b) Final shapes observed in the diagrams above, the colors indicate corresponding regions in those diagrams. The dark green lines represent their final $h(t)$ schematically. One may notice the ascending order from the centered parachute to the off-centered slipper. This order is always preserved when λ is fixed. The parachute snaking and complex tumbling are considered as intermediate states of the other three steady states.

This region serves as an intermediate stage situated between the near-centered slipper and off-centered slipper configurations. Remarkably, this yellow region also experiences a shift towards the left-hand side. This broad-scale evolution of phase boundaries with variations in both λ and W (or C_n) may potentially suggest an alternative representation of the results wherein both parameters could be varied according to a certain functional dependence $p(\lambda,C_n)$. In essence, instead of altering both parameters independently, a single combined variation might yield a more suitable description of the system's behavior.

The outcomes depicted in Fig. 4 should be interpreted as a projection of the results from the (λ,C_n) parameter plane onto a singular line $p(\lambda,C_n)$. This function is postulated to exhibit monotonic behavior with respect to both λ and C_n, adhering to the conditions $\partial p/\partial\lambda > 0$ and $\partial p/\partial C_n < 0$. It's pertinent to note that the degree of confinement C_n doesn't qualitatively alter the dynamics of vesicles within the considered parameter range ($1 \leq \lambda \leq 10$, $1/10 \leq C_n \leq \frac{1}{2}$).

The simulation findings presented in Fig. 3, along with their robustness emphasized in Fig. 4, collectively highlight the distinct nature of the branches $h_f^{(OC)}$ and $h_f^{(NC)}$. For the branch $h_f^{(OC)}$, a saddle-node bifurcation is exhibited, where the red line (representing $h_f^{(OC)}$ in Fig. 4) and the dashed line demarcating the border of the red zone characterize its stable and unstable branches, respectively. Conversely, the branch $h_f^{(NC)}$ emerges from a pitchfork bifurcation stemming from the parachute solution. In Fig. 4, the cyan line signifies $h_f^{(NC)}$, and its negative counterpart $-h_f^{(NC)}$ both constitute stable branches, while $h\equiv 0$ represents the unstable branch.

In this context, we endeavor to estimate the degenerate parameter $p(\lambda,C_n)$. The underlying concept is to identify the points of bifurcation onset, represented by the critical viscosity contrasts $\lambda_c^{(NC)}$ and $\lambda_c^{(OC)}$, as functions of the confinement C_n.

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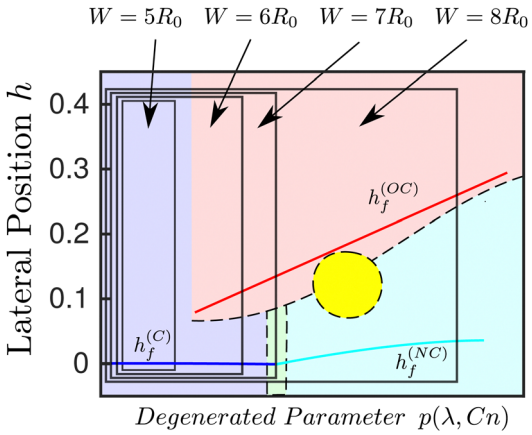


Fig. 4 A schematic plot showing an attempt to combine both C_n and λ into a single parameter p. The black frames plotted with black solid lines (indicated by arrows) are mappings of the diagram from h *vs.* p back to h *vs.* λ plot. One may notice that each frame contains a similar diagram corresponding to simulation results (in Fig. 3a). The final positions of the three steady shapes are plotted in blue ($h_f^{(C)}$ for parachute), cyan ($h_f^{(NC)}$, near-centered slipper), and red ($h_f^{(OC)}$, off-centered slipper).

Subsequently, we scale the viscosity contrast λ relative to these pivotal critical values.

Within the confines of the bounded Poiseuille flow, the presence of channel walls tends to exert a centralizing influence on vesicles.[52] Conversely, elevating the viscosity contrast λ prompts migration away from the center.[26,52] Consequently, when the viscosity contrast reaches a certain threshold, these opposing forces reach equilibrium, culminating in an off-centered position. As the channel width diminishes (equivalently, the confinement C_n increases), the wall-induced lift effect becomes more pronounced, requiring a higher viscosity contrast to achieve a balanced off-centered position (as illustrated in Fig. 3a).

For branches that are near-centered, the critical viscosity contrast $\lambda_c^{(NC)}$ scales proportionally to ${C_n}^2$ at low confinement ($C_n < 0.22$), while in more confined channels, it scales as ${C_n}^3$, as showcased in the left panel of Fig. 5a. Similarly, the critical viscosity contrast for off-centered branches $\lambda_c^{(OC)}$ exhibits scaling behavior as C_n at low confinement ($C_n < 0.29$), and shifts to scaling as ${C_n}^2$ under higher confinement conditions (as demonstrated in the right panel of Fig. 5a).

Consequently, we introduce two degenerate parameters by recalibrating the viscosity contrast λ as:

$$p^{(NC)} = \frac{\lambda}{\lambda_c^{(NC)}}, \quad p^{(OC)} = \frac{\lambda}{\lambda_c^{(OC)}}. \tag{14}$$

As both $\lambda_c^{(NC)}$ and $\lambda_c^{(OC)}$ exhibit an increase with rising C_n (as depicted in Fig. 5a), the equations presented above naturally fulfill the conditions $\partial p/\partial\lambda > 0$ and $\partial p/\partial C_n < 0$. We have illustrated all the gathered data within the plane delineated by the final lateral position and the degenerate parameter p, as visualized in Fig. 5b. This depiction, grounded in the framework of a degenerate parameter, results in a compelling collapse of data points. This observation suggests an underlying, concealed correlation between the viscosity contrast λ and the confinement C_n. A parallel scaling phenomenon in the context of shear flow has been documented by Thiebaud *et al.*[53]

An intriguing query arises regarding the existence of two distinct degenerate parameters, $p^{(NC)}$ and $p^{(OC)}$. The origin of the near-centered branch is attributed to a pitchfork bifurcation, while the off-centered branch corresponds to a saddle-node bifurcation. Consequently, these two branches remain disconnected, offering no inherent reason for $p^{(NC)}$ and $p^{(OC)}$ to adhere to the same scaling behavior. It is our intention to delve further into this matter, drawing from direct analytical considerations, in order to establish a more foundational understanding of the underlying principles.

3.3 Effects of capillary number

The capillary number Ca is a measure of membrane deformability. The increase of flow velocity or the decrease of membrane bending modulus are typical ways of increasing Ca. We investigated the same parameter set (h_0,λ) as in Section 3.2 but with different values of Ca. The results are shown in Fig. 6. We found no topological changes for the off-centered slipper region (which has a final position $h_f^{(OC)}$), when comparing the results to those of Fig. 4. However, when Ca is decreased from 80 to 8 and then to 4, we observed a significant shrinkage and even collapse (when $W \gtrapprox 7R_0$) of the parachute region, see Fig. 6. Here we show the results for $W = 10R_0$ as an example, while similar behaviors are observed when $W \gtrapprox 7R_0$. This observation was also reported in Kaoui *et al.*[15] The fact that this occurs for a capillary number of order unity is indicative of the competition between the hydrodynamic stress and the bending resistance. Note that for $Ca \lessapprox 8$, the time needed to reach final shapes becomes long. For example, when $Ca = 4$, $\lambda = 4$ and $h_0 = 0.4$, it took about $10^3\tau_c$ to reach the final position and final shape. The dashed line with cyan color in Fig. 6 represents the terminal position (near-centered slipper) for $Ca = 4$.

An insightful comparison can be drawn between the dynamics of two scenarios obtained at two extreme capillary numbers. Let's consider the points A1 and A2 with coordinates (Ca,λ,h_0) = (4,6,0.1) and (4,6,0.05), respectively. Additionally, let's examine the points B1 and B2 with coordinates (80,6,0.1) and (80,6,0.05) (as displayed in Fig. 6). When $Ca = 80$, the dynamics of both B1 and B2 exhibit a complex tumbling motion, whereas for $Ca = 4$ (points A1 and A2), a stable near-centered slipper shape is achieved.

In Fig. 7, the lateral position curves [$h(t)$] along with their corresponding shapes are plotted for the initial $1000\tau_c$. For the case of $Ca = 4$, simulations with both $h_0 = 0.05$ and 0.1 result in tumbling with slight deformation, and their lateral positions gradually decrease until they reach the final state. Additional simulations conducted with initial positions ranging from $h_0 = 0.04$ to 0.01 indicate that $h(t)$ continues to decline until it eventually reaches $h_f^{(NC)}$—the final position for the near-centered slipper shape. However, it's worth noting that the entire tumbling process lasts for an extended period, approximately $t \sim O(10^4\tau_c)$.

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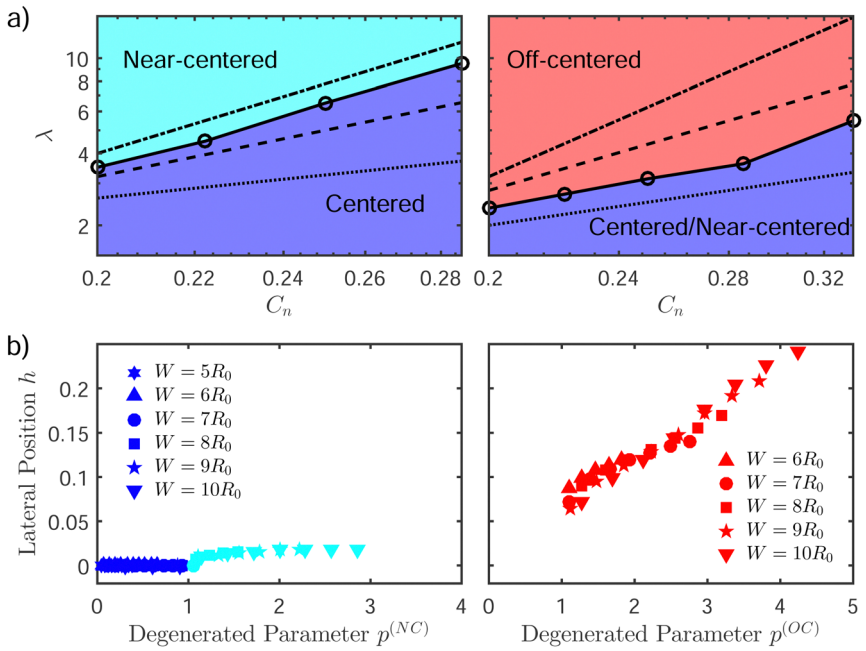


Fig. 5 (a) Phase diagrams in the plane of confinement C_n and viscosity contrast λ. The circles represent the boundaries of final positions, and the solid lines are a guide for the eyes. We also show straight lines corresponding to slope = 1 (dotted line), slope = 2 (dashed line), and slope = 3 (dash-dotted line). (b) Evolution of final positions against degenerated parameter $p^{(NC)}$ (left panel) and $p^{(OC)}$ (right panel). The blue, cyan, and red colors indicate the centered, near-centered, and off-centered branches, respectively.

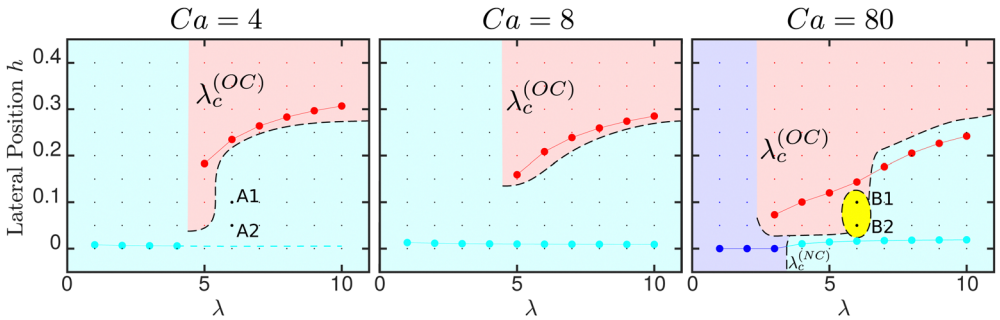


Fig. 6 The evolution of final position h_f with viscosity contrast for Ca = 4, 8 and 80. The colors and final positions share the same definition as in Fig. 3 and 4. In all these simulations, C_n is fixed at 0.2 ($W = 10R_0$). It is confirmed that the off-centered slipper is robust and exists for all explored capillary numbers. The parachute shape is not preferred when Ca is small. When $\lambda = 1$, our simulation results are consistent with the results from Kaoui *et al.*[16]

In contrast, for Ca = 80, the complex tumbling motion (as observed in cases B1 and B2) persists. It's intriguing to observe that the shape depicted in Fig. 7 for Ca = 80, which can be termed a "trilobe shape" as suggested by ref. 7 and 17, manifests even in the absence of cytoskeleton elasticity, signifying its robust nature. This observation underscores that this specific shape remains independent of the intricacies of the underlying structure. The region where this type of motion predominates is indicated by the yellow zones in Fig. 3.

3.4 Impacts on rheology

The objective of this section is to comprehensively analyze the rheological characteristics of the suspension, spanning from highly diluted to relatively dense states. In the realm of dilute conditions, our aim is to establish a connection between the solution branches discussed in the preceding sections and the resulting rheological behavior.

Let's revisit the fundamental findings from the preceding sections. For confinement values within the range of 0.2 to 0.4 and viscosity contrasts spanning $1 \lessapprox \lambda \lessapprox 10$, it was revealed in Sections 3.1–3.3 that a prominent effect of elevated viscosity contrast is the emergence of an off-centered slipper shape (with its corresponding final lateral position denoted as $h_f^{(OC)}$). Additionally, the exploration of a broad spectrum of capillary numbers (Ca) ranging from 4 to 80 unveiled that the off-centered slipper region exhibits the most extensive basin of attraction for viscosity contrasts within the range of $4 \lessapprox \lambda \lessapprox 7$. Notably, this viscosity contrast

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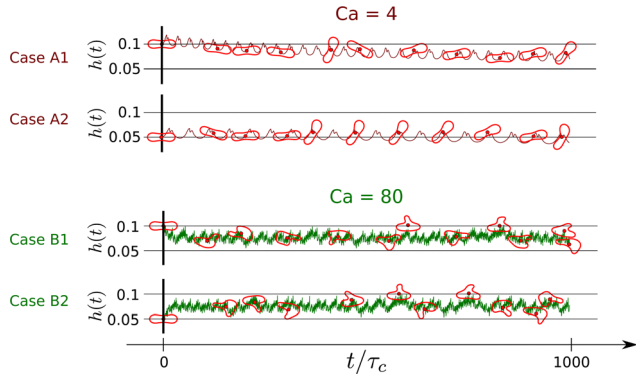


Fig. 7 Emergence of a complex attractor upon an increase of *Ca*: in all 4 cases, λ is fixed at 6. When *Ca* = 4, both A1 and A2 yield $h(t)$ curves with a downward drift, which could imply *a priori* that this tumbling motion could be unstable. However, long time simulation confirms convergence of this solution. For *Ca* = 80, $h(t)$ corresponding to B1 and B2 always oscillate in a complex fashion between 0.05 and 0.1, illustrating the existence of a complex attractor, with a chaotic-like signature.

range corresponds to the values typically observed in healthy RBCs.

The area concentration of a vesicle suspension is defined as $\phi = n{\cdot}\pi R_0{}^2/(LW)$, where n represents the number of vesicles within the suspension. The effective viscosity of the vesicle suspension, denoted as η_{eff}, is ascertained from the linear correlation between the average flux and the pressure gradient,[54] as expressed by:

$$\eta_{\text{eff}}{\cdot}Q = \eta_{\text{out}}{\cdot}Q^0 \tag{15}$$

where Q and Q^0 represent the average flux of the vesicle suspension and the pure fluid (with viscosity η_{out}), respectively, under the same pressure gradient. The average flux Q and Q^0 are defined as

$$Q = \frac{1}{L}\int u_x \mathrm{d}x\mathrm{d}y,\ Q^0 = \bar{u}_x^0 W, \tag{16}$$

where $\bar{u}_x{}^0$ is the mean velocity of the imposed flow. We recall that η_{out} is the viscosity of external fluid. The intrinsic (or normalized) viscosity, $[\eta]$ – which evaluates the contribution of the suspended entities (vesicles in this study) to the viscosity of a solution – can be defined as

$$[\eta] = \lim_{\phi\to 0}\frac{\eta_{\text{eff}} - \eta_{\text{out}}}{\eta_{\text{out}}\phi} \tag{17}$$

The total velocity $\mathbf{u}$ is composed of the imposed flow $\mathbf{u}^0$ [the first term of the right-hand-side of eqn (6)] and the induced flow $\mathbf{u}^{(\text{ind})}$ [the terms represented by integrals in eqn (6)]. One can express the normalized viscosity as a function of the induced flow and the areal concentration. We straightforwardly find

$$[\eta] \approx \frac{\eta_{\text{eff}} - \eta_{\text{out}}}{\eta_{\text{out}}\phi} = -\frac{\int u_x^{(\text{ind})}\mathrm{d}x\mathrm{d}y}{(\bar{u}_x^0 WL + \int u_x^{(\text{ind})}\mathrm{d}x\mathrm{d}y)\phi} = -\frac{[u_x^{(\text{ind})}]}{1 + [u_x^{(\text{ind})}]\phi} \approx -[u_x^{(\text{ind})}], \tag{18}$$

where the last equality is valid in the dilute regime. $[u_x{}^{(\text{ind})}]$ is the normalized mean induced velocity (along x) defined as

$$[u_x^{(\text{ind})}] = \frac{\left\langle \int u_x^{(\text{ind})}\mathrm{d}x\mathrm{d}y \right\rangle}{\bar{u}_x^0 \cdot n\pi R_0{}^2} = \frac{\left\langle \sum_{i=1}^{n}\int u_{x_i}^{(\text{ind})}\mathrm{d}x\mathrm{d}y \right\rangle}{\bar{u}_x^0 \cdot n\pi R_0{}^2} \tag{19}$$

In the provided equation, denoted as eqn (19), the notation $\langle\cdot\rangle$ represents the operation of time averaging, which can be omitted in the context of steady shapes. The term $n\pi R_0{}^2$ corresponds to the aggregate area covered by all vesicles immersed in the flow, and $u_{x_i}^{(\text{ind})}$ signifies the flow velocity that is induced due to the presence of the ith vesicle.

Eqn (19) indicates that the influence of ϕ is implicitly embedded within $u_x{}^{(\text{ind})}$. Consequently, as the value of ϕ decreases, both the numerator and denominator of the expression tend towards zero. By utilizing the boundary integral formula as presented in eqn (6), it becomes evident that $[u_x{}^{(\text{ind})}]$ contains direct information about the shape of the vesicle. This shape, in turn, is contingent upon various dimensionless parameters such as λ, C_n, and *Ca*. In our evaluation, we numerically compute $[u_x{}^{(\text{ind})}]$ using an equidistant spatial mesh size $\Delta x = 0.1R_0$.

Building upon the concept of the normalized mean induced velocity $[u_x{}^{(\text{ind})}]$ as defined in eqn (19), the local normalized induced velocity can be formulated as:

$$U_x^{(\text{ind})}(\mathbf{r}) = \frac{u_x^{(\text{ind})}(\mathbf{r})}{\bar{u}_x^0 \cdot \phi} \tag{20}$$

The above discussion means that $[u_x{}^{(\text{ind})}]$ is approximately equal to $-[\eta]$ when the suspension is dilute [see eqn (18)].

We plot the $U_x{}^{(\text{ind})}$ fields for two distinct steady states: a centered parachute (as shown in Fig. 2c) and a slipper at an off-centered position (as shown in Fig. 2b) respectively, fixing parameters at *Ca* = 80, λ = 6 and $W = 7R_0$. A quick glance to that plot shows that the figure which exhibits a darker blue color (Fig. 8a) would provide a higher viscosity, since this is related to the negative average value of the induced field.

Fig. 9 depicts the behaviors of h_{f} and $[\eta]$ as functions of λ, encompassing both centered and off-centered branches at *Ca* = 80. For *Ca* = 8 and 4, aside from an increase in the absolute magnitude of $[\eta]$, no significant qualitative modifications were observed in their respective diagrams. These observed trends can be summarized into three distinct classes:

(i) For the centered parachute shape, the normalized viscosity $[\eta]$ exhibits minimal sensitivity to variations in λ, while its absolute value demonstrates an approximate proportionality to C_n.

(ii) For the off-centered slipper shapes (illustrated by the red curves in Fig. 9), the final position $h_{\text{f}}^{(\text{OC})}$ positively correlates with the value of $[\eta]$. This behavior is attributed to the equilibrium between the lift arising from the wall[52,55–61] and the lateral migration towards the off-center direction, which occurs when the viscosity contrast is high.[26,52] Simultaneously, the increase in $h_{\text{f}}^{(\text{OC})}$ coincides with a rise in effective viscosity,

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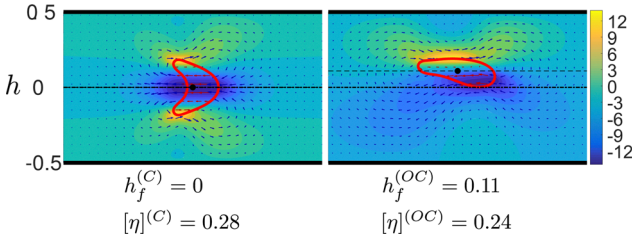


Fig. 8 The total induced flow field for vesicles possessing different final shapes with the same parameter set [(Ca,λ,C_n) = (80,6,2/7)]. The color map indicates the component of the normalized local induced velocity field along x, $U_x^{(ind)}$ [see eqn (20)], which has its mean value approximately equal to $-[\eta]$. The off-centered slipper has its normalized viscosity $[\eta]^{(OC)} \approx 0.24$ (right panel), and is about 15% smaller than its counterpart for the centered parachute (left panel, $[\eta]^{(C)} \approx 0.28$).

a phenomenon known as the Fahraeus-Lindqvist effect.[62] This effect serves as the primary reason for the interdependence between $h_f^{(OC)}$ and $[\eta]^{(OC)}$.

(iii) Regarding the near-centered slipper shapes (portrayed by the blue curves in Fig. 9, where h_f has non-zero values), an increase in λ prompts a minor reduction in $[\eta]$. This trend arises because the near-centered slipper shape, in comparison to the parachute shape, exhibits a smaller lateral extent in the vertical direction and demonstrates a tank-treading motion. These attributes lead to a softer impedance against the flow, contributing to a slight decrease in effective viscosity. This tendency is further accentuated with higher values of λ.

As the width W (or C_n) increases, the off-centered slipper shape becomes more pronouncedly off-centered, subsequently causing a greater lag with respect to the imposed flow. This lag results in enhanced dissipation, ultimately leading to a crossover point where the off-centered solution exhibits a higher viscosity in comparison to the parachute solution. This observation underscores the intricate relationship between vesicle shapes and their dynamic interactions with the flow field.

3.5 Investigation of rheology for higher concentrations

In this section, our focus shifts to the analysis of rheological properties at higher concentrations. For this purpose, we employ the lattice-Boltzmann method (LBM), which offers greater efficiency, particularly in high-concentration scenarios. Unlike the boundary integral method (BIM), where the computational complexity scales as $O(n^2)$ for a suspension of n vesicles, LBM exhibits a more favorable scaling of $O(n)$.[44] Additionally, the LBM provides relatively accurate results. We have previously validated the LBM by comparing its outcomes with BIM results for various test cases.[44] In this section, we extend this comparison by examining the results of both methods for higher vesicle concentrations.

In order to facilitate this comparison, we conduct systematic studies using different initial configurations. It has been reported that under linear shear flow,[53] vesicle suspensions organize into linear files, leading to non-monotonic behavior in the normalized viscosity $[\eta]$. This behavior arises as a consequence of confinement and the wall migration effect experienced by the vesicles. Given that physiological flows often resemble Poiseuille flows, it is crucial to explore how the rheological properties are influenced in this case.

In Fig. 10, we present the normalized viscosity $[\eta]$ and the effective viscosity η_{eff} as functions of concentration ϕ for various viscosity contrasts (λ = 1 and 5). The fixed parameters include Ca = 80, $W = 7R_0$, and $L = 8W$. The number of vesicles, denoted by n, ranges from 2 to 56, corresponding to concentrations $\phi \in [0.016,0.448]$. The simulations commence with well-relaxed vesicle shapes and random positions within the channel. This analysis allows us to gain insights into the influence of vesicle concentration on the rheological behavior, especially with respect to different viscosity contrasts.

Fig. 10a and b depict the normalized viscosity ($[\eta]$) as a function of ϕ, demonstrating the consistency between the LBM and BIM results (as shown in Fig. 10a). A notable observation emerges: $[\eta]$ consistently decreases with ϕ up to a concentration

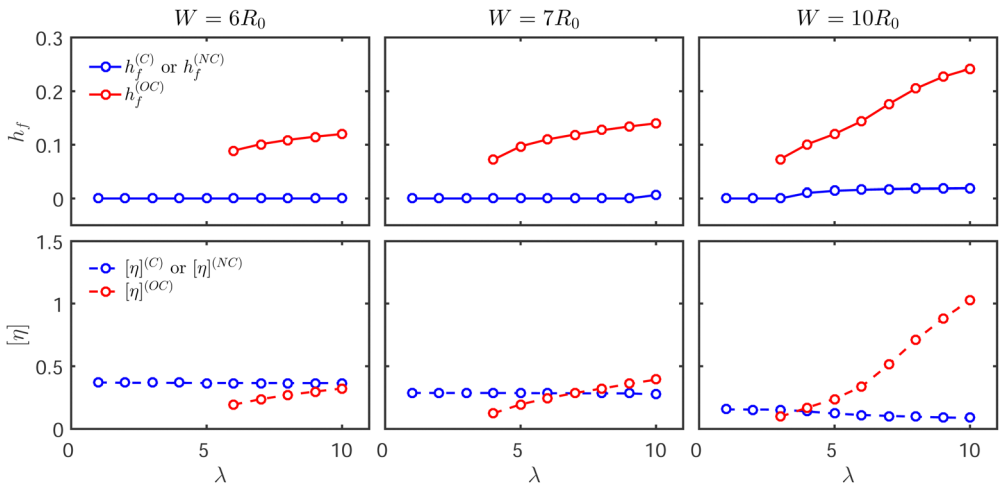


Fig. 9 Final position h_f and normalized viscosity $[\eta]$ as a function of viscosity contrast λ for different values of channel width (or confinement). The capillary number is fixed at Ca = 80.

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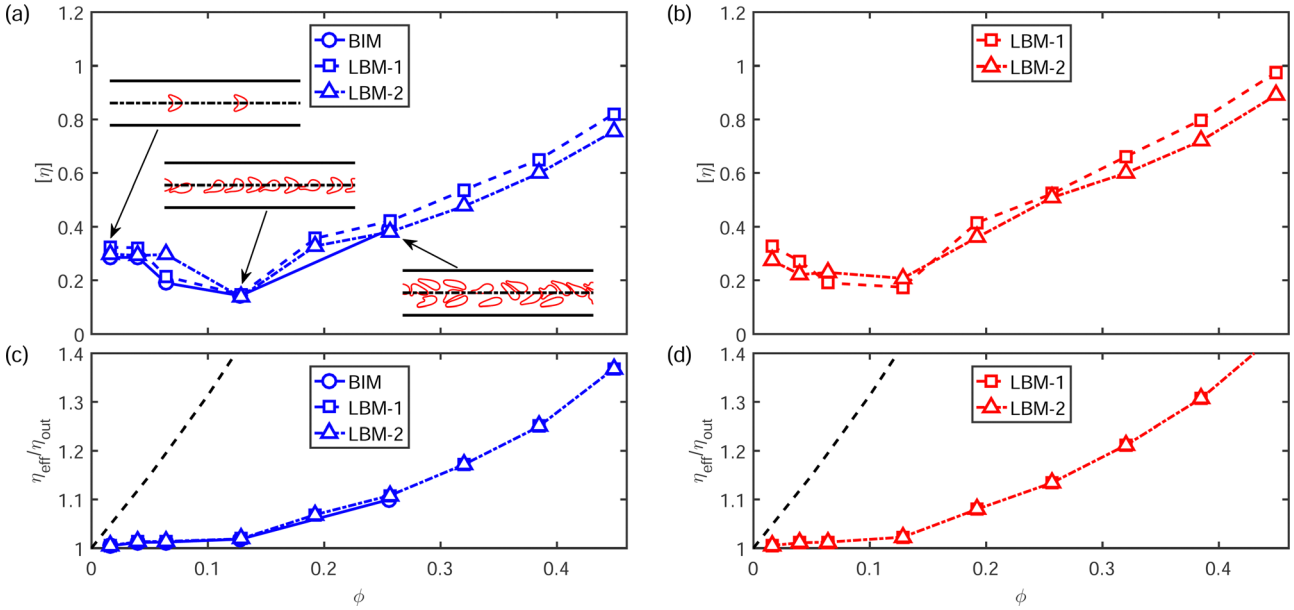


Fig. 10 The normalized viscosity $[\eta]$ (a and b) and the effective viscosity η_{eff} (c and d) as a function of concentration ϕ for different values of viscosity contrast λ (λ = 1 for the left column, and λ = 5 for the right column). Here BIM and LBM represents the results using boundary integral method and lattice-Boltzmann method, respectively. Different random initial configurations of vesicles are studied when using LBM, denoted by 1 and 2. The insets in (a) are snapshots of vesicles at ϕ = 0.016, 0.128, and 0.256. The dashed line in the lower panel is the relative viscosity of blood in macrocirculation according to Pries *et al.*[63] Here $W = 7R_0$ and Ca = 80.

of approximately $\phi \lessapprox 0.13$; further increasing the value of ϕ leading to a monotonic increase of $[\eta]$. This trend persists across different initial vesicle configurations (labeled as 1 and 2 in Fig. 10), revealing a robust behavior.

This intriguing behavior is elucidated through intuitive explanations. It's noteworthy that the reduction in $[\eta]$ occurs within a concentration range aligned with microcirculation values, typically ranging from 5% to 20%.[64,65] In this context, the system appears to strategically minimize dissipation to enhance the flow rate of cells, facilitating more effective oxygen transport.

To comprehend this behavior, we refer to our previous findings for a single vesicle, which remain valid in the low ϕ regime. For λ = 1, we observed that a single vesicle migrates to the center, adopting a centered parachute shape (refer to Fig. 1a). In the sufficiently dilute regime, vesicles align into a single file, positioning themselves at the center or its vicinity. This arrangement resembles the behavior of isolated vesicles, indicating weak hydrodynamic interactions among them. As vesicles arrange themselves, recirculation zones appear in the spaces between them (see the upper panel of Fig. 11). The inter-distance between vesicles is approximately $10R_0$, in agreement with the findings of Aouane *et al.*[66] The presence of these recirculation zones introduces backflow, which leads to a decrease in the average flux Q.

As hematocrit increases, signifying the addition of new cells, the incoming cell integrates into the single-file formation, thereby reducing the intensity of the recirculation zone (middle and lower panels of Fig. 11). This reduces dissipation per cell, and subsequently diminishes the associated backflow due to

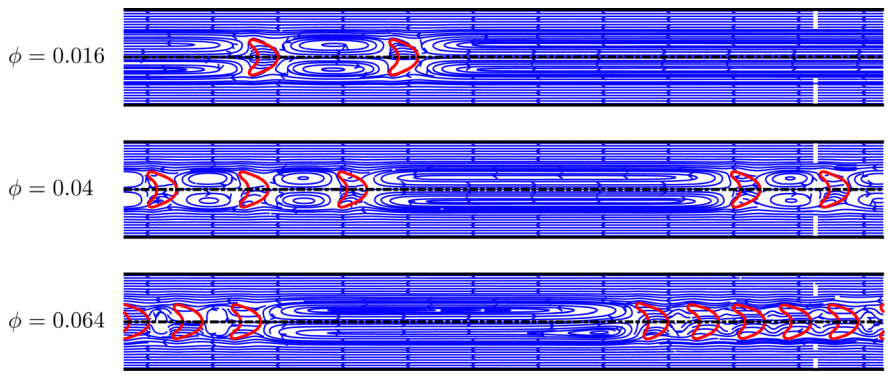


Fig. 11 Streamlines (blue lines with arrows) in a comoving frame for results of LBM-1 in Fig. 10a. Vesicles are denoted with red lines. The corresponding concentrations are ϕ = 0.016 (upper panel), ϕ = 0.04 (middle panel), and ϕ = 0.064 (lower panel). Here $W = 7R_0$, λ = 1 and Ca = 80.

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each recirculation zone. Consequently, $[\eta]$ decreases with ϕ within a small range of ϕ values. As ϕ is further increased, some central cells are expelled towards the periphery, where they encounter increased resistance to the flow. This results in an increase of $[\eta]$ with ϕ (Fig. 10a and b).

In contrast, the effective viscosity η_{eff} consistently increases with ϕ (Fig. 10c and d). The non-monotonic behavior of $[\eta]$ has a significant impact on the behavior of η_{eff}, causing its mild increase with ϕ. This increase is much slower than what is typically observed in macrocirculation. In macrocirculation, blood viscosity increases by about 2-fold at $\phi = 0.25$, whereas in microcirculation, it only experiences a modest 1.1-fold increase. This subtle cell organization, contributing to a wide cell-free layer, plays a pivotal role in this phenomenon.

It is intriguing to explore whether the increase in viscosity is predominantly due to the cell-free layer or the organization of cells. For instance, in the study by Thiebaud *et al.*,[53] under linear shear flow, $[\eta]$ decreases with ϕ at low ϕ, then increases with ϕ after a single central file becomes saturated, leading to expulsion of some cells from the center. Further increasing ϕ then causes $[\eta]$ to decrease again, as the cells organize into two symmetric files. This organization resulted in a very mild increase (or even a plateau) of the effective viscosity η_{eff} with ϕ.[53] Our findings suggest that in microcirculation, increasing hematocrit has a relatively minor impact on viscosity, facilitating a more efficient oxygen supply. This phenomenon likely contributes to the performance enhancement observed in athletes upon hematocrit increase, driven by a minimal viscosity increase with ϕ in microcirculation.

4 Conclusions

In this study, we have conducted a comprehensive numerical investigation into the dynamical behavior of vesicles within a channel flow subjected to Poiseuille conditions. Several branches of solutions have been identified, including centered parachute and slipper shapes, snaking parachute, off-centered slipper configurations, and complex trilobe dynamics. The phase diagram, characterized by viscosity contrast and initial position, exhibits intricate complexity. Furthermore, we have demonstrated that the combined effects of viscosity contrast and confinement can be encapsulated into a single degenerate parameter.

Our exploration underscores the complexity of dynamics even within the simplest 2D model, featuring an inextensible membrane with resistance against bending. Importantly, we have established that the trilobe shape is not exclusive to membrane cytoskeleton interactions, refuting the assertion made in Mauer *et al.*[7] While enriching the model by incorporating shear elasticity in a full 3D context might introduce additional complexities, the modes we have identified represent robust features that do not necessitate additional model components.

We have extended our analysis to rheological properties at various concentrations. These rheological behaviors are closely connected to the spatial organization of vesicles. A remarkable observation is the decrease of the normalized viscosity $[\eta]$ in the relatively low concentration range (Fig. 10), a phenomenon not restricted solely to linear shear flow,[53] but also present in pipe flow scenarios, which more accurately model blood circulation. This decrease can be attributed to the spatial organization of vesicles. In dilute suspensions, the cells tend to adopt a single-cell solution (slipper shape) that minimizes dissipation. The induced flow field within the gap between adjacent cells plays a role akin to screening. Notably, the viscosity decrease occurs within a range of approximately 0 to 15%, aligning with the hematocrit range observed in microcirculation. In a sense, cells self-organize to minimize the viscosity increase associated with higher hematocrit, optimizing the efficiency of oxygen transport. Correspondingly, the effective viscosity exhibits a slow increase (or even a plateau tendency; refer to Fig. 10), ultimately leading to a decrease in normalized viscosity.

It remains an intriguing future endeavor to investigate whether the observed decrease in $[\eta]$ is a robust characteristic that extends to 3D systems. The study conducted by Thiebaud *et al.*[53] within a linear shear flow context demonstrated the same rheological trend as observed in 2D. However, those investigations were confined to dilute suspensions (approximately 5%). Further exploration of spatial organizations in 3D and their implications on rheology, particularly at higher hematocrit levels, represents an engaging avenue for future research.

Another perspective concerns the effects of spatially non-homogeneous confinement on blood flow. On the inner surface of a blood vessel, endothelial cells form wavy surfaces.[67,68] Experimental and numerical studies have shown that wavy vessels result in oscillations of cell morphology,[69,70] and emergence of an off-centered attractor.[71] In pathological situations, the formation of stenosis leads to sudden changes in vessel confinement as well.[72] While passing through constrictions, cells/vesicles undergo large deformation[73–75] and lateral migration.[76,77] Future studies considering spatially non-homogeneous vessels are essential in order to understand cell dynamics and blood rheology under realistic conditions.

Author contributions

Zhe Gou: methodology (theory and simulations), software, investigation, writing – original draft, review and editing. Hengdi Zhang: methodology (theory and simulations), software, investigation, writing – original draft, review and editing. Abdessamad Nait-Ouhra: conceptualization, methodology (theory and simulations), review and editing. Mehdi Abbasi: investigation, review and editing. Alexander Farutin: conceptualization, methodology (theory and simulations), software, review and editing. Chaouqi Misbah: conceptualization, methodology (theory and simulations), review and editing.

Conflicts of interest

There are no conflicts to declare.

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